



Society of Petroleum Engineers

SPE-226652-MS

Single Trip Multifrac Operation with Sliding Sleeves on a Subsea Template in the North Sea

Alexander Letichevskii and Jan Håvard Jøranson, Equinor; Mike Vanee, Stig Horve, James Furukawa, and Patrick O'Hare, NCS Multistage

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/226652-MS](https://doi.org/10.2118/226652-MS)

This paper was prepared for presentation at the SPE International Hydraulic Fracturing Technology Conference and Exhibition held in Muscat, Sultanate of Oman, 23 - 25 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

As global energy systems evolve, operators are increasingly compelled to extract value from complex, tight reservoirs, assets once considered uneconomical or too technically challenging to develop. The North Sea, with its mature infrastructure and evolving field redevelopment strategies, represents a frontier where such innovations can create outsized impact.

This paper documents a successful collaboration to implement a single-trip multistage hydraulic fracturing system using multicycle sliding sleeves on a subsea template in the Gullfaks South field. It is the first time this technology has been applied in this specific configuration in the North Sea, and it offers a transformative approach to stimulating tight, compartmentalized formations like the Lunde reservoir. The goal is to share a proven methodology that allows operators to achieve:

- Enhanced recovery factors from tight formations previously deemed uneconomic
- Lower operational risk when compared to other hydraulic fracturing techniques where proppant under-displacement is required
- Operational scalability for future subsea developments with jack-up, semi-sub rigs, or floating rigs.

Introduction

Hydraulic fracturing has evolved significantly in offshore environments, driven by the need for improved efficiency, cost-effectiveness, and productivity from low-permeability reservoirs. Multistage fracturing (MSF) and single-trip multistage fracturing (STMSF) techniques have emerged to address these needs, enhancing stimulation operations.

The work builds upon lessons learned from comparable applications in the Danish sector (Maersk, SPE-205331), Clair Field (BP, SPE-216632), Oseberg and Statfjord Fields (Equinor, OTC-35380) and Shell's Gulf of Mexico campaign (SPE-220794), reinforcing an industry-wide shift toward high-efficiency, simplified stimulation techniques that reduce offshore time and expand the limits of what is technically feasible in cases where proppant under-displacement is necessary.

By sharing technical data, operational planning models, and lessons learned, this paper aims to contribute meaningfully to industry knowledge in a domain where continued innovation is critical to energy security and low-carbon resource development.

Reservoir and Field Context

The Gullfaks Sør (South) field, located in block 34/10 in the northern North Sea, is a satellite field developed by remotely operated subsea wells which are tied back to one of the platforms on the Gullfaks main field.

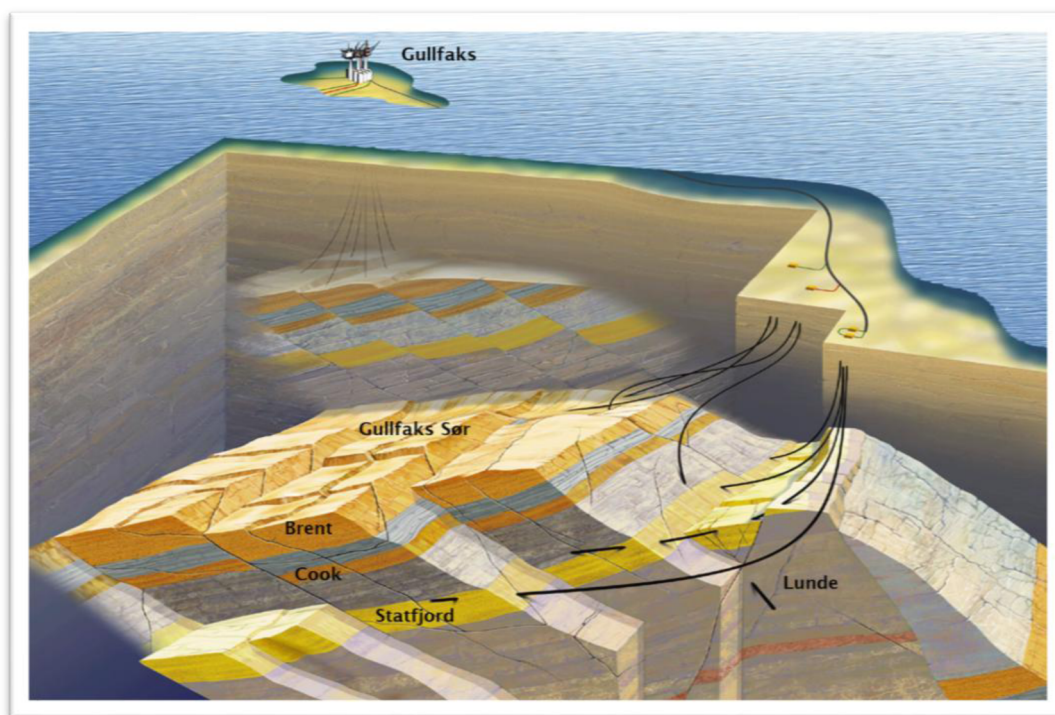


Figure 1—Gullfaks Sør (South) field

The field development started in 1998 targeting the conventional reserves concentrated in Brent, Cook, and Statfjord formations. However, reserves located in the deepest part of the field, Lunde formation, were not developed due to previously being considered uneconomic. Lunde formation is a thick package with a gross thickness of approximately 700 meters of fluvial deposits. The formation is mainly oil-filled, with a gas cap down to about 3240 meters TVD MSL. It is a compartmentalized formation which consists of channel sands (varying thickness) interbedded with floodplain deposits as shown on Fig 2. The average permeability in Lunde formation is in a range of ~0.9-1.9 mD, and sand porosity ranges from 14–17%. A large portion of the total field in place volumes are in the tight Lunde formation. Despite the large volume of hydrocarbons in place, the recovery factor is very low. Only the upper part of Lunde with the best reservoir properties has been produced so far. Even the long, highly deviated cased and perforated wells showed disappointing rates, and rapid rate decline. To improve productivity and recovery from low-permeability Lunde formation Company has chosen to drill a horizontal well with multistage proppant fracturing to connect isolated sand bodies through shale barriers and enhance reservoir contact.

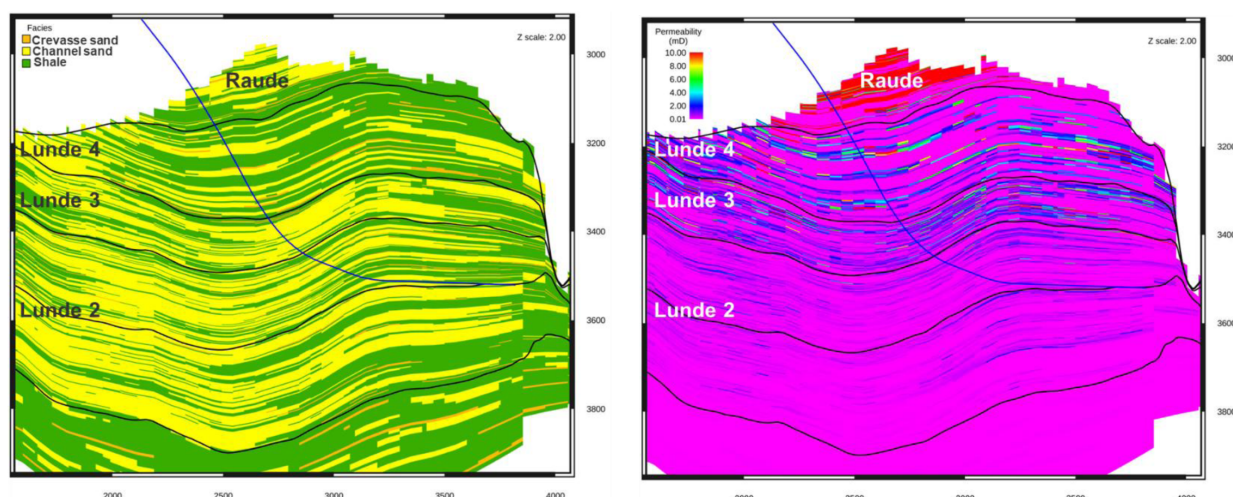


Figure 2—a) Distribution of the lithology and b) distribution of the permeability

Based on the economic evaluation of the project, a minimum of 5 successful frac stages are required to have a robust economy and secure oil and gas production. Therefore, a well with 7 frac stages was planned to account for any operational issues with pumping the frac jobs or operating the sleeves. The strategy was to drill a well through the entire Lunde formation. But setting TD at a safe distance to the OWC. Most of the stages (four) would be placed in the deeper part of the reservoir with the lowest permeability – Lunde 2.1 and Lunde 2.2, Fig.3. Furthermore, it was decided to place one fracture in the middle part of the reservoir with slightly better reservoir properties, Lunde 3, to improve overall recovery from this part of the formation. In addition, it was decided to stimulate the upper part of the reservoir, which is gas-filled, with two frac stages to achieve an in-situ gas lift for the oil zones. In the beginning the well should produce naturally due to high overpressure in the deeper part of the reservoir. Pore pressure in Lunde 2.1 and Lunde 2.2 is approximately 1.54 sg. To increase overall reservoir drainage, it was decided to drill the well in the direction of minimum horizontal stress, North-South, to induce transverse fractures as shown in Fig. 4.

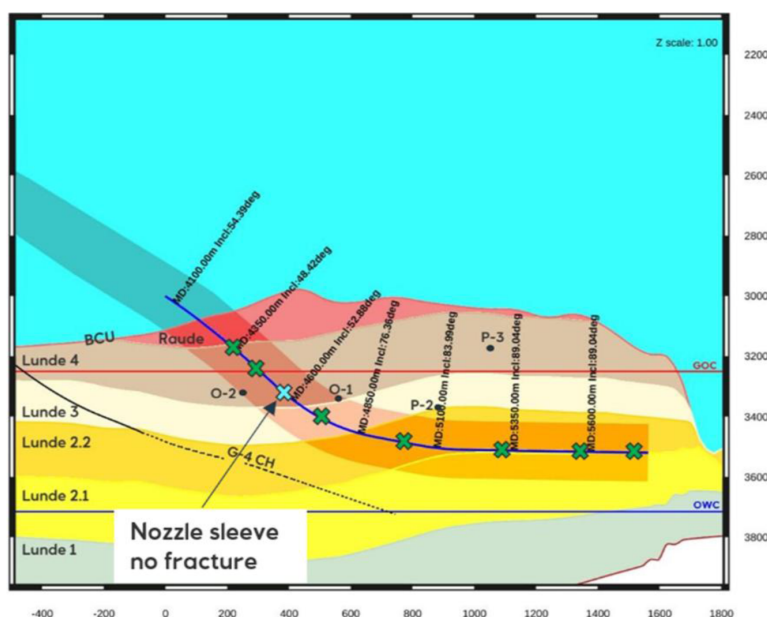


Figure 3—Gullfaks Sør G1 AH Well Sleeve Placement with Planned Fracture Radius shown as shading around the wellbore

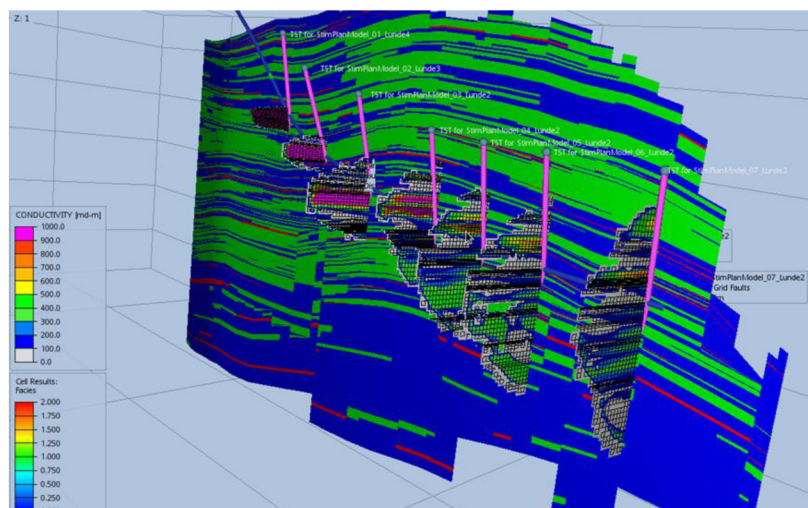


Figure 4—Simulated fracture geometry imported into the reservoir simulator

Fracturing design focused on maximizing the height coverage to connect as many sand bodies as possible. Therefore, it was decided to pump fairly massive frac jobs of more than 450 klbs of proppant and more than 500 m³ of frac fluid at high pumping rates of 25-30 bpm. The delayed 35-45# crosslinked fluid was used as a frac fluid throughout the entire pumping job. The main reason for choosing the high polymer loading was to achieve sufficient fluid stability during the long frac treatment and taking into account a high reservoir temperature - 135 deg C for the deepest zone. All of this resulted in fractures with fracture half-length of about 100 m and fracture height of about 100 m. Based on the proppant selection study, 20/40 intermediate strength proppant has been chosen. To maintain sufficient fracture conductivity, all stages were planned with a low PAD percentage and maximum proppant concentration of 10 ppa. It was decided to pump the entire frac treatments with resin coated proppant (with resin coating of 4.7 weight %) to prevent proppant flowback during the well production phase. To ensure a good connection between wellbore and fractures, Company decided to perform all frac treatments with proppant under-displacement technique, where 1-2 m³ of the proppant slurry (with final concentration) will be left at the end of the pumping job inside the string and across the frac sleeve after each treatment. Therefore, it is required to reverse out this under-displacement prior to moving to the next stage to ensure a clean string.

Well G-1 was drilled in 2023 and stimulated in May 2024. A drill pipe deployed system was selected for multistage fracturing of the well. Fracturing campaign was performed after the lower completion was installed but before the upper completion. Since this was the first fractured well in the field, a mini-frac was pumped in most of the stages. Drill pipe conveyance enables operational efficiency to manipulate frac sleeves, even for under displaced treatments, and it is possible deliver 2 fracture treatments per day, significantly reducing operational time and cost as compared to previous methodologies.

This paper outlines key learnings from applying a frac-through-pipe methodology on a subsea template, along with frac sleeve technology with operational insights and recommended next steps.

System Selection for Offshore Fracturing Operations

Operator has selected drill pipe-deployed systems as the primary method for hydraulic fracturing operations across its global offshore assets. This decision is the result of a comprehensive evaluation process that considered multiple operational, logistical, and technical factors. Below is a summary of key considerations that guided the selection process.

Drill Pipe vs. Coil Tubing Deployment

Advantages of Drill Pipe–Deployed vs Coil Tubing–Deployed Fracturing Systems.

- Main advantage: very efficient clean-up when proppant under-displacement is used or in case of a screenout.
- Proppant remains inside the pipe, resulting in negligible risk of stuck pipe (ref. SPE 212902-MS)
- Having "live annulus" (static column) enables real-time downhole pressure monitoring during the frac treatments
- Coiled tubing (CT) operations generally require the use of a double flapper check valve assembly to prevent wellbore fluids from coming back to the surface – this prevents the possibility of reverse circulating up by pumping down the annulus and taking returns to surface.
- Use of Standard Equipment: Utilizes industry-standard drill pipe and slick pipe (non-upset), offering operational familiarity and availability.
- Lower Mobilization Costs: Coil tubing is more expensive to mobilize offshore and typically requires more personnel during rig-up and operations, especially when executed through the derrick.
- Reduced System Friction compared to pumping through CT: Lower friction allows for higher fracture rates and pressures.
- BOP, drilling systems, X-mas tree are not exposed to proppant since all clean-out operations are performed in reversing out mode.
- Greater Mechanical Force: More downhole force is available for manipulating frac sleeves and conducting cleanup operations.
- CT would demand more footprint on offshore installations.

Advantages of Coil Tubing–Deployed Fracturing Systems.

- Pipe Movement: While drill pipe can be moved under pressure through the annular bags, movement is limited to sections of pipe that must be threaded and unthreaded to move. CT is continuous and can move in and out of the well under high pressure utilizing a stripper assembly.
- Improved Operational Efficiency Between Stages: Less time is required for pressure testing and frac head make-up between stages. However, recent operational efficiencies utilizing drill pipe have been streamlined to reduce this challenge.
- Parallel Operations on Fixed Platforms: Allows fracturing to be conducted in parallel with drilling operations from the main derrick. This is highly complex and dependent on platform design, space available, and upfront planning. (BP)(SPE 191431)
- Efficiency in Multi-Well Operations: Coil tubing may be more efficient for multi-well fracturing from the same platform, albeit with limitations on system rate and proppant containment.
- Reduced Friction via Annular Fracturing: Annular frac techniques are often employed to mitigate system friction. The risk of stuck pipe is a key consideration in the economics of an operation. (Aker BP) (SPE-212902-MS)

Drill Pipe Deployment vs. Plug & Perforate (P&P) or Ball Drop (BD) Systems

- Screen-Out Recovery: In the event of a screen-out using drill pipe (or coil tubing), the slurry can be reverse circulated, or, in the case of fluid loss from the open stage, the frac sleeve can be closed and proppant reversed to surface for cleanup. Operations can typically resume within a few hours. In contrast, P&P or BD systems require coil tubing or jointed pipe intervention to clean up the screen-out zone from surface to the point of blockage, which will effect the cost and operational efficiency.

- Fracture Workflow Flexibility: Drill pipe-deployed frac sleeves allow for stepwise operations including breakdown tests, mini-fracs, and main fracs—while maintaining a closed wellbore between each injection stage. It also allows out of sequence fracturing operations.
- Controlled Proppant Placement: Supports under-displacement to retain a proppant-filled conduit between the reservoir and sleeve ports.
- Fluid management and efficiency: rather than "bull heading" fluids, a drill pipe conveyed system provides the ability to forward circulate fluids down via drill pipe by taking returns up the annulus. This enables cost efficiency if swapping from lower specific gravity fluids for fracturing to heavier specific gravity brines to maintain well control while moving in-between frac stages. In addition, it allows for lower pad volumes if required.
- Long-Term Reservoir Management: Frac sleeves allow for lifetime well management, including selective shut-off of stages producing water or unwanted gas.

Multistage Hydraulic Fracturing Sleeve Description and Qualification. The multistage hydraulic fracturing sleeves are deployed as an integral component of the liner string within the lower completion system. Engineered specifically for offshore applications, the sleeves feature a high-torque, high-tensile strength body and have been qualified to ISO 14998 V3 standards to ensure reliable performance under challenging wellbore conditions.

Each sleeve is actuated using a dedicated shifting mechanism, which may include a tubing-conveyed frac isolation assembly (FIA) or a wireline-conveyed stroker. The sleeve assembly consists of four primary components: a top sub, an outer barrel, a bottom sub, and an inner sleeve. During installation, the inner sleeve is retained in the closed position by brass shear pins. During the stimulation operation, a FIA is used to locate and actuate the sleeve. Upon proper positioning and anchoring within the inner barrel, the FIA applies a hydraulic piston force of 10 MPa to shear the brass pins, displacing the inner sleeve downward and exposing the frac ports located in the top sub.

To enable pre-stimulation diagnostics and post-stimulation interventions, the system allows for multiple open and close cycles. After the initial opening event, a collet-based latching mechanism secures the sleeve in either the open or closed position. Repeated actuation is achieved by overcoming the 6,000 lbf latch force via hydraulic pressure from the FIA piston or mechanical set-down weight to open the sleeve, and a tensile force (applied while anchoring into the inner sleeve) to return it to the closed position.

The sleeve includes a dedicated sealing system designed to provide pressure integrity between internal and external annular pressure regimes during both installation and stimulation operations. Qualification testing was conducted in accordance with ISO 14998 Annex A to validate performance under the combined effects of temperature, internal/external pressure, and axial loads expected during deployment, stimulation, and production phases.

Additionally, to validate the functionality of the sealing and mechanical systems over multiple open/close cycles, the sleeve underwent testing in accordance with ISO 14998 Annex D. This testing confirmed the sleeve's ability to maintain pressure integrity and mechanical reliability through repeated actuation events across the well's operational life.



Figure 5—Multistage Hydraulic Fracturing Sleeve showing four major components: (a) closed fracturing sleeve, (b) open fracturing sleeve

Nozzle Sleeve. One of the frac sleeves was equipped with carbide nozzles in place of standard production ports to provide production flexibility based on downhole reservoir evaluation. This modification was introduced to accommodate uncertainties in the Raude formation's petrophysical properties, sand strength, and reservoir connectivity - factors that would be assessed during drilling. Depending on the final reservoir characterization, the Raude zone could be either co-produced with the Lunde or produced separately. The nozzle sleeve was intended to restrict Raude's contribution in case the reservoir properties were superior. After drilling the well, it was decided not to place the Nozzle Sleeve in a Raude section since it can be connected with the fracture initiated in Lunde 4 formation. Instead, it was placed between frac stages 5 and 6 to allow influx from about 400m of open hole section in middle part of Lunde formation.

Frac Isolation Assembly. The Frac Isolation Assembly (FIA) is a tool designed to shift sleeves open or closed, allowing access to the reservoir. For this application, the FIA has been specifically engineered for thru-tubing fracturing operations on jointed pipe. The FIA operates in four distinct modes to facilitate sleeve actuation: Run-In-Hole (RIH), Pull-Out-Of-Hole (POOH), Compression Set, and Tension Set. These modes are cycled through continuously using a "jay" system by reciprocating the work string.

The frac sleeve depth is identified in the POOH mode via a string weight overpull, which occurs when the FIA's collet engages with a profile within the sleeve. Once positioned, set-down weight engages slips with the sleeve's inner barrel and compresses the isolation element to form a seal. Differential pressure is then applied to the seal, shearing pins within the sleeve and shifting it open to expose the frac ports. While set in the sleeve and isolating the casing below, a frac can be pumped down the work string and out through a frac wedge, which directs flow from inside the tubing to the exposed ports. After the frac is placed, picking up the work string cycles the FIA to the next mode, allowing the bi-directional slips to engage with the sleeve and close it. From there, the pattern repeats, and the tool can be cycled back into POOH mode to locate the next sleeve.

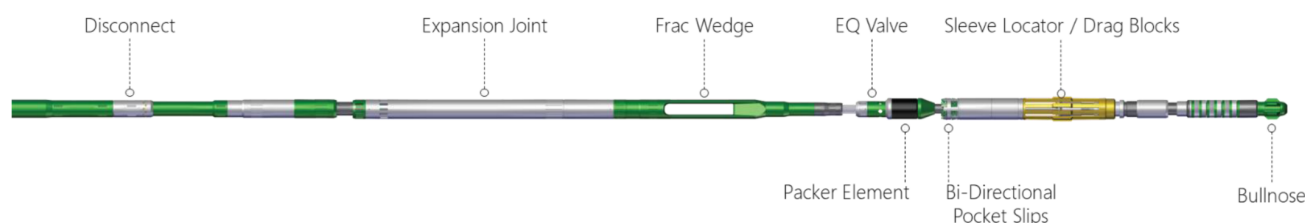


Figure 6—Frac Isolation Assembly (FIA) Components

Design Improvements for Drill Pipe Fracturing. The main components of the FIA that required design modifications to improve functionality for drill-pipe fracturing are described below:

Locating Collet

To enable accurate FIA positioning, the assembly incorporates a mechanical sleeve locator consisting of a drag collet with locating dogs. These dogs provide a noticeable overpull force when the FIA is POOH and

engaged with a sleeve profile. The original design produced an overpull of approximately 5,000 lbf, which was insufficient given the total weight of the work string. As such, the collet and locating components were redesigned to increase the overpull force to between 15,000–20,000 lbf. This ensures a more substantial and discernible change in the weight indicator on the surface.

Expansion Joint

The Expansion Joint is a critical FIA component designed to absorb axial movement resulting from sleeve engagement. In POOH mode, when the mechanical sleeve locator is engaged in a sleeve profile, tensile force is applied to the tubing until the locator is released. The Expansion Joint absorbs this movement, preventing premature cycling of the FIA into the next mode within the continuous jay pattern.

In jointed pipe applications, the Expansion Joint also plays an essential role in compensating for tubing movement caused by thermal contraction. In the G1 well, which had a static bottom-hole temperature (BHT) of 135°C, the introduction of cool frac fluid (as low as 15–20°C) was expected to result in tubing contraction of up to 4 meters. Since a slick joint must be positioned within the blowout preventer (BOP) during stimulation of each zone, it was crucial that tubing movement that occurs below the BOP maintains the tool stay in compression set mode and ensure the slick joint remained within the BOP stack.

Unlike coiled tubing operations, where movement through an annular stripper can easily be facilitated during stimulation to accommodate thermal contraction, jointed pipe operations face limitations in string movement. The outward-stroking capability of the Expansion Joint ensures that this movement is absorbed, preventing downhole displacement of the FIA, as the string remains anchored by slips at surface.

Frac Wedge

Although not a new concept, the frac wedge plays an important role in transitioning the stimulation slurry from tubing-conveyed flow to annular flow, directing it through the sleeve ports and into the formation. With proposed stimulation rates as high as 30 bbl/min and a planned sand tonnage of over 450,000 lbs per stage, the design of the frac wedge was critical to the overall system performance. The geometry of the wedge, its position relative to the sleeve ports, and the hardness of the selected material were all optimized to combat erosion caused by high-velocity, turbulent flow during this transition. These design considerations helped ensure reliable slurry delivery, minimized tool wear, and maintained flow path integrity over the course of multiple high-rate, high-volume frac stages.

Previous work, and a full-scale erosion test was completed in 2023 (ref Shell SPE-220794) where 6.8MM lbs. 20/40 mesh ceramic sand was pumped at 35bbl/min. The results were favorable with minimal damage to the frac wedge.

Subsea Fracturing via Drill Pipe: Operational Planning

Key Considerations

For the Gullfaks Sør G template, the use of the Jack-Up Rig Askeladden in water depths of 134m presents a unique operational scenario: performing stimulation operations from a jack-up rig using a subsea wellhead and BOP stack. This configuration introduces new challenges compared to traditional stimulations conducted from jack-ups using tubing-conveyed FIA systems and surface BOPs. However, it also offers a valuable opportunity to vet and adapt this technology for future deepwater applications using semi-submersibles or drill ships.

Primary Barrier

Kill weight brine considered as a primary barrier during single-trip multistage fracturing (STMSF) thru-drill pipe. Based on the predicted and measured pore pressure during the well drilling phase, a 1.56 specific gravity (sg) CaCl₂/CaBr₂ kill weight brine was chosen to ensure well control. Fluid management was a key part of the pre-job planning to ensure safe pipe movement and maintain well control throughout the operation. The plan called for multiple fluid swaps between heavy weight brine, used while moving drill

pipe, and a 1.02 sg freshwater-based frac fluid for stimulation. In the end of each frac treatment the string will be filled with light weight frac fluid. Hence, these swaps were necessary to manage hydrostatic pressure while moving from one operational step to another or from stage to stage. The use of drill pipe allowed for forward circulation of frac treatment down the string, taking returns of heavy weight brine up the annulus, and reverse circulation to clean up under displace stimulation slurry, enabling controlled and efficient fluid exchanges, together with preventing from bullheading or loss of expensive heavy weight brine during the fracturing operations.

Four key operational stages were identified where fluid losses or gains could occur: Opening the Sleeve, Breakdown Test, Mini-Frac, and Main Frac. In all cases, the primary barrier was the 1.56 sg $\text{CaCl}_2/\text{CaBr}_2$ brine in the annulus, which had to be maintained to ensure well control.

Table 1—Operational sequence and associated risks

Operation	Objective	Risk	Mitigation
1. Perform breakdown test	Confirm injectivity and sleeve functionality	Losses opening sleeve or after breakdown	Close sleeve Top up annulus
2. Space out & RU frac stand	Avoid tool joint across BOP	Damage of annular element	Verification of the space-out by different parties
3. Connect vessel, test lines	Prepare for fracturing ops	Vessel collision Release of energy	DP system, Bridging document, procedures and barriers
4. Perform drills	Crew training	Communication	Repeating training until everyone is comfortable Repeat training for every new crew
5. Perform diagnostic tests (mini frac, SRT)	Lower the risk of screenout and optimize the main job	Release of energy NWB pressure out	PT of surface lines and barriers Crew trainings
6. Reversing out LG after the diagnostic test	Establish primary barrier – brine in DP	Unable to close sleeve Gas to surface	Set NCS BHA above the sleeve Reversing out with const BHP
7. Perform main frac	Stimulate the reservoir, i.e. increase production,	Screenout Release of energy	Diagnostic tests + field experience Fluid QAQC on the vessel
8. Reversing out XLG w/ proppant after the main	Establish primary barrier – brine in DP	Unable to close sleeve Plugged DP Gas to surface	Crew trainings and contingency flow charts Set NCS BHA above the sleeve Reversing out with const BHP
9. Rack back frac stand and POOH to next frac sleeve			

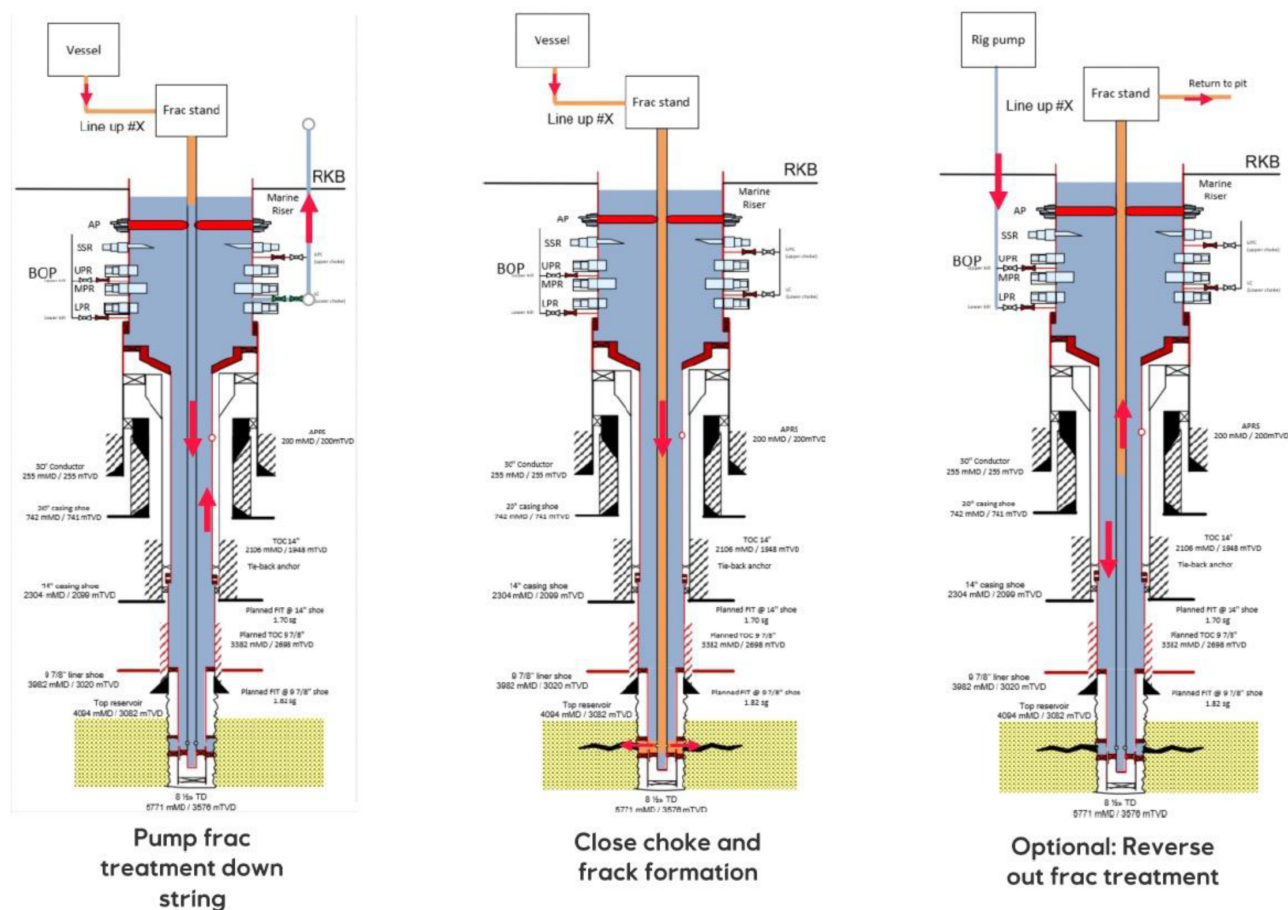


Figure 7—Main frac planned sequence (heavy weight brine in blue and light weight frac fluid in orange)

Slick Joint Positioning, Annular Integrity, and Frac Head Height Constraints

One of the primary risks identified with the subsea BOP configuration is the potential for incorrect spacing of the 5.0" drill pipe (slick joint) within the BOP stack while the FIA is set in an open frac sleeve during stimulation. The slick joint is critical for ensuring that no tool joint passes through the annular element of the BOP. This is essential because the annular bag must be closed around the smooth body of the drill pipe to establish pressure isolation for circulating, shifting frac sleeves, and performing stimulation operations.

Range 3 drill pipe was selected due to its extended length, which provides the flexibility needed to accommodate the vertical movement required for cycling the FIA through its different operating modes: locating sleeves, shifting sleeves open, closing sleeves, and circulating fluids when positioned either above or below the sleeve. These operations rely on precise vertical pipe movement, and any misalignment could have serious consequences. If a tool joint is inadvertently pulled through the closed annular, it significantly increases the risk of:

- Damage or accelerated wear of the annular elements,
- Element replacement, which adds non-productive time,
- Debris generation, where rubber fragments could plug surface lines,
- Or in the worst case, failure of the annular to contain pressure as planned.

In addition to the annular integrity concerns, operational constraints also arise from the height of the frac head relative to the rig floor. As the subsea BOP sits approximately 190 m below the rotary table, surface-

level frac head positioning must be coordinated with slick joint spacing. If the slick joint placement requires the frac head to be positioned too high or low, issues can arise such as:

- Straining or limiting the flexibility of frac hoses, which can lead to equipment wear or restricted movement during pumping.
- Inability to operate the FIA to close the sleeve after the frac operations or set it above as a contingency measure.

This interdependency between slick joint depth and frac head height necessitated careful modeling and preplanning to ensure both annular integrity and surface equipment compatibility were maintained throughout the operation.

Corrective Operations: Short Trips. If slick joint placement was incorrect, a short trip was required to reposition the joint—pulling pipe out of hole and rerunning to ensure the slick section landed correctly inside the BOP. While the procedure itself is straightforward, it adds significant time and cost, particularly with the rig costs and frac vessel standby costs while waiting for the next sleeve. To mitigate this, a proactive approach was taken to model space-out scenarios and planned joint selection in advance.

Mitigation Strategy: Pre-Job Modeling and Planning Tools

Space-Out Calculator. To avoid tool joints landing in the BOP and minimize the need for corrective short trips, a pre-job Excel-based model was developed. This tool integrated the liner tally (defining sleeve and BOP depths) with the drill pipe tally to predict the position of the FIA shifting tool and ensure a slick joint would land cleanly within the BOP window. The model allowed for proactive adjustment of the string design using pup joints to fine-tune spacing.

This modeling was performed in two stages:

- Pre-job planning: A detailed subsea simulation iterated through joint combinations—swapping long joints and inserting pup joints—to ensure a slick joint would align with the BOP on the first run.
- Real-time surface checks: During operations, minor adjustments were made at surface by shuffling the upper joints or adding short pups to maintain the frac head within a 2 m stick-up window at the rig floor.

This two-tiered approach enabled accurate space-out across all seven sleeves, eliminating the need for short trips and ensuring safe annular sealing.

½" liner with mechanical open hole (OH) isolation packers, which had the max OD of 8 ¼". In addition, drilling ensured low dog leg severity, less than 3 deg/30m. The 5 ½" 20 lb/ft 13Cr80 liner with 8 sliding sleeves and 14 OH isolation packers were set from 3,909.8 m to 5,721.24 m. The lower completion was set in oil-based mud and then the well was displaced to heavy weight brine during the clean out run. The deepest 4 sleeves were located in the horizontal section with inclination close to 90 deg while the shallowest ones were in the build-up section with inclination between 50 and 80 deg.

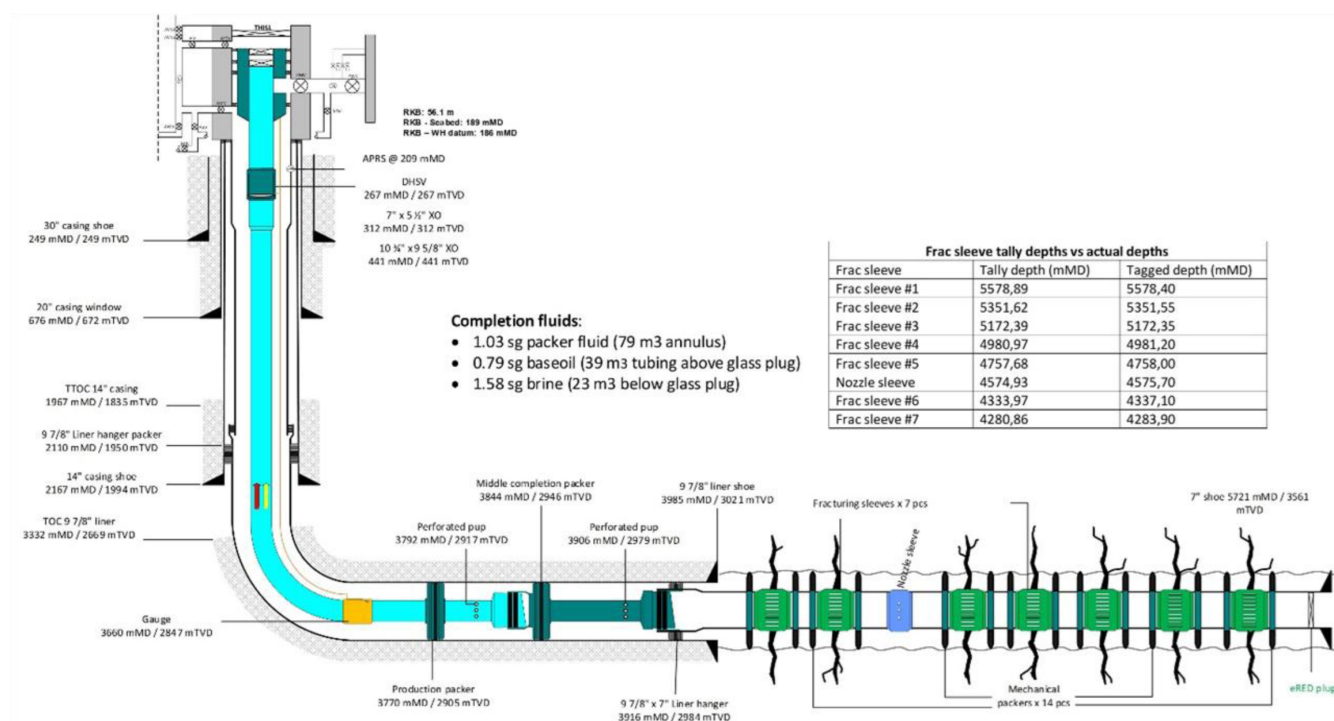


Figure 9—Completion Diagram

Fracturing Operation

Overview

The frac operation began in May 2024 once a favorable working window became available from previous weather delays. The FIA was already made up and RIH on a workstring consisting of 5.0" 19.5 lb/ft S-135 drill pipe and 3.5" 9.20 lb/ft L-80 tubing to be readily available for stimulation operations to begin. With the FIA in hole and below the sleeve, the desired sleeve was identified with a measured overpull of string weight in the sleeve locate profile. The sub-surface BOP spaceout was verified based off of sleeve locate profile depth, and the corresponding slick joint pipe body spacing above and below the annular bags to be stripped through. The FIA was then positioned in the sleeve and RIH to Compression Set, engaging the slips and the element within the sleeve. Pressure was then applied, and the sleeve was shifted open. With the sleeve confirmed open, stimulation operations were commenced as desired. The 8 sleeves in the well (7 frac + 1 nozzle) were successfully shifted open and closed for a total of 22 sleeve cycles.

Sleeves 1, 3, 4, 5, 7 and 8 were shifted open prior to the following stimulation operations: breakdown test, mini-frac and main frac. Sleeve 2 only required the initial breakdown test and then progressed straight into the main frac. Sleeve 6 (nozzle sleeve) was shifted open and immediately closed, as no injection was required on that sleeve.

The breakdown tests were performed utilizing the cement unit with the kill weight brine as the fluid. Each mini-frac and main-frac were completed with the stimulation vessel with the fluid as described below.

After each stimulation operation, the sleeves were closed before proceeding with the next operation or to the next target sleeve.

Each stage used a 35-45 ppg delayed cross-linked gel system. The base fluid was fresh water with 2 % KCl. 20/40 mesh ceramic sand was planned with 100 mesh loaded on the vessel as optional. Proppant concentration was stepped 1 → 10 lb/gal. Total pumped: ~ 2.6MM lb (1.2MM kg) across 7 stages. Individual sleeves received 133 k – 466 k lb; Pump rate was 20 - 30 bpm (4.8 m³ min) for all mini-fracs and main treatments. All stages were pumped according to the design without screenouts.

Stage 4 Formation Breakdown Challenges: The sleeve was initially opened, and a breakdown test was conducted using the cement unit. Afterward, the sleeve was closed and pressure tested. Preparations for the mini-frac followed. When crosslinked gel was injected into the formation, a near wellbore tortuosity resulted in very high pressure, and the electronic shut-down of the pumps was reached. After several attempts to pump the mini-frac, it was decided to temporarily suspend operations on Stage 4 and proceed to Stage 5. After successfully fracturing all subsequent stages, Stage 4 was revisited. This time a linear gel was pumped instead of a crosslinked gel with positive results. A mini-frac followed by a main frac was performed, and the stage was successfully stimulated with the placement of 466,051 lbs (214 tons) of proppant.

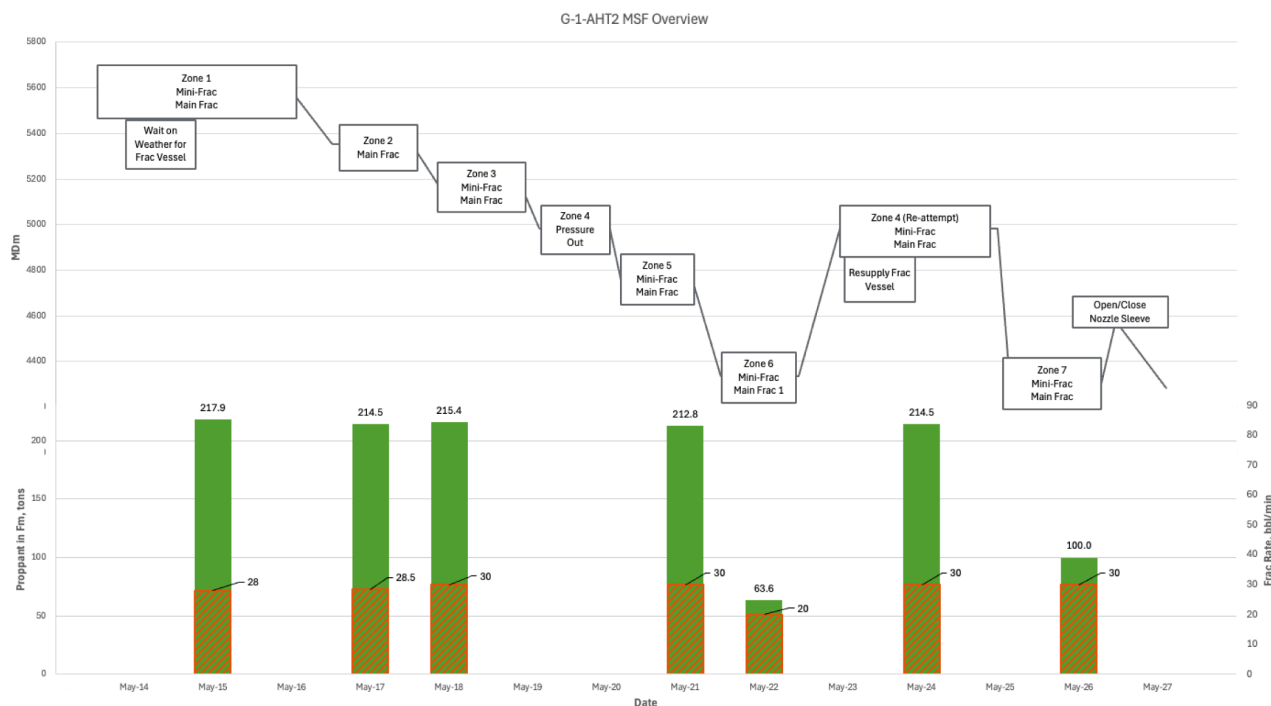


Figure 10—G-1-AHT2 Multistage Fracturing (MSF) Overview

Post Frac Operations and Production Results

Clean up wellbore

Following the completion of frac operations, the FIA was run below Sleeve 1. A reverse circulation clean-up was performed to remove any residual proppant or debris from the wellbore, ensuring unobstructed access for subsequent wireline operations.

Middle Completion

To provide an ISO V-0 barrier and isolate the reservoir during the installation of the upper completion and subsea Christmas tree, a middle completion was installed. This consisted of a glass plug and a middle completion packer.

Upper completion

A standard upper completion was installed, comprising a production packer, pressure/temperature gauge, and subsurface safety valve.

Opening frac sleeves for production

Following the installation of the upper completion, the jack-up rig was demobilized and moved to the next drilling location. A light well intervention (LWI) vessel was subsequently mobilized to open the frac sleeves and the nozzle sleeve in preparation for production. Using electric wireline equipped with a tractor, stoker tool, and anchor, all sleeves were successfully opened as planned.

The production forecast, based on assumed reservoir properties, was 390 Sm³/D of oil plus 600 000 Sm³/D of gas. As seen from the figure xx, the initial production was more than twice the forecast. Another concern in the planning, was proppant flowback. In reality, the total proppant production in the first 6 months was about 150 kg, in other words not a concern!

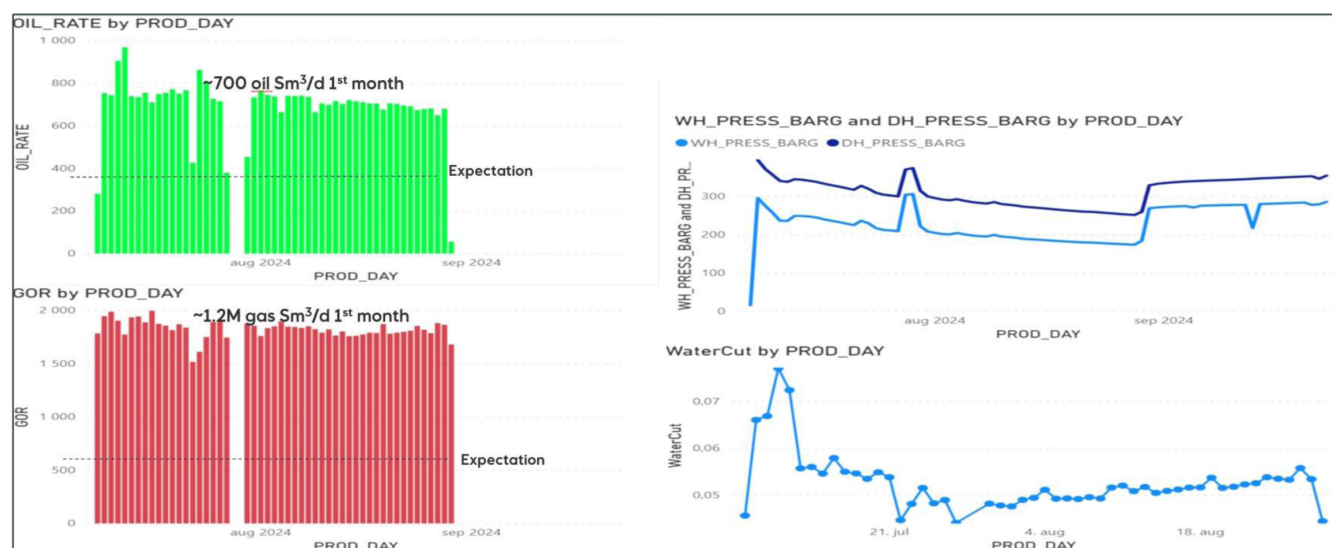


Figure 11—Production results during the first two month after the frac operations

Summary and Lessons Learned

The stimulation campaign was executed safely, and largely as planned. All seven stages were shifted and treated successfully, with no screen-outs, pressure anomalies, or mechanical failures encountered. Several stages were preceded by diagnostic testing—mini-fracs and breakdowns—which provided useful data and helped confirm the reservoir response before committing to full-volume treatments. This added a layer of confidence to the execution and allowed for minor adjustments as needed. The system allowed sleeves to be manipulated in and out of sequence, which introduced flexibility in how the operation could be planned and adjusted.

Operational Efficiency during the fracturing phase

Operational efficiency during the fracturing phase was generally in line with the pre-job plan. The only meaningful downtime came from weather-related delays, which are typical for offshore North Sea operations. Despite those interruptions, the team stayed on schedule and delivered the full stimulation scope without major issue. The vessel only re-loaded once with proppant since the vessel is equipped with a reverse osmosis (RO) system to generate fresh water.

Pipe handling was clean throughout, with minimal short trips, and the system performed as designed. One change made was to increase the number of pup joints available for the next job to provide more flexibility

for spacing out the drill pipe. The ability to utilize drill pipe instead of CT in the execution of the stimulation program enabled under-displacement of proppant across the target zones. This helped ensure strong near-wellbore connectivity.

The implementation of the drill pipe deployed FIA with the sliding sleeve technology demonstrates significant operational efficiency. When compared to existing methodologies which take 2-3 days per stage, achieving 2 frac treatments in a 24hr timeframe is a reasonable estimate for execution with this technology. (ref. SPE-205331-MS)

FIA Locking Feature

After the operation was complete, a locking mechanism was conceived to prevent premature activation of the FIA while running into the well. This feature temporarily disables the FIA during deployment, allowing for pull tests and safe movement through the casing without risk of triggering the tool's functionality.

The locking feature also prevents the FIA from cycling through its operational modes while RIH, which is critical when transitioning into the 5.5" liner. Without this control, there is a risk of setting the packer prematurely—potentially damaging the slips or the isolation element. With the locking mechanism in place, the tool remains in a neutral, locked state until a defined trigger is applied at depth to enable normal operation. While it was primarily intended to improve control during deployment on fixed platforms, it also serves as a proactive measure for future floating rig operations, where string movement and heave-induced cycling could otherwise trigger the tool prematurely.

Sleeve Closure Challenge and Mitigation Strategy

A consistent minor operational challenge was observed in 5 of 7 stimulation stages involving difficulty in executing the initial tension-set to close the frac sleeves. Following under-displacement of the slurry, attempts to pull tension and shift the sleeve closed resulted in the bottomhole assembly (BHA) sliding uphole rather than achieving the intended sleeve closure. In each case, the operational procedure was adapted by setting the packer, reverse-circulating the remaining slurry from the drill pipe and returning to the sleeve to close on the second attempt. Successful closure was confirmed by setting the FIA below the sleeve of interest and pressure testing.

Subsequent post-job surface testing was unable to consistently reproduce the problem under controlled conditions. Although a definitive root cause was not established, the introduction of an added drag assembly during lab testing effectively mitigated the problem. As a preventive measure for future jobs, a revised drag assembly will be integrated into the FIA to increase drag during tension-set operations. Ongoing surveillance and tool performance monitoring along with any additional improvements are planned to ensure continued reliability.

Well Clean-up Post Frac

Prior to the post-frac trip to total depth (TD), bottomhole gauge data confirmed that the FIA elastomer element maintained pressure integrity throughout the stimulation. This indicates that the tool was effectively isolating pressure during the treatment. However, following the clean-up run, surface inspection revealed damage to the elastomer element. The root cause of this damage is still under investigation.

Despite the post-run damage, the FIA successfully enabled reverse circulation during the clean-up operation. This functionality helped maintain a clean annulus and prevented proppant from reaching surface flow lines. The clean-out procedure involved pumping high-viscosity gel pills followed by a venturi junk basket run. Both the gel pill returns and the junk basket came back clean, suggesting that this step may be unnecessary in future operations.

Over the first six months of production, approximately 150 kg of solids were recovered at the production facility. This observation supports effective proppant transport and stage clean-up during the initial flowback period.

Production Results

Production results exceeded expectations. The performance was strong enough to justify planning five additional wells using the same approach. Based on the overall outcome, this method—frac-through-completion on drill pipe—appears to be a viable and repeatable solution for subsea environments.

Acknowledgements

The authors would like to thank the partners in the Gullfaks Sør licence, Equinor Energy ASA, Petoro and OMV, and NCS Multistage Management for their permission to publish this paper. The authors would also like to acknowledge the contributions and hard work from both the Equinor team and the NCS Multistage teams. We extend our gratitude to our internal team members, whose hard work and collaborative spirit was instrumental in achieving this successful outcome.

Henrik Wolter, Sigurd Eikenes, Daniel Cabrera Jordan, Gro Selås Kallevik, Olav Solberg Eikrem - Equinor

Brad Smith, Cullen Taylor, Jonathan McGregor, Kurtis Goller, Lyden DeCecco, Warren Krenzler, Jason West, Barry Gillespie – NCS Multistage

List of Abbreviations

BHA	Bottomhole Assembly
BHT	Bottomhole Temperature
BD	Ball Drop
BOP	Blowout Preventer
CT	Coiled Tubing
FIA	Frac Isolation Assembly
FM	Formation
ISIP	Instantaneous Shut-In Pressure
LWI	Light Well Intervention
MSF	Multistage Fracturing
P&P	Plug and Perforate
POOH	Pull Out of Hole
RIH	Run In Hole
SG	Specific Gravity
STMSF	Single-Trip Multistage Fracturing
STOIP	Stock Tank Oil Initially In Place
V0	ISO V-0 (Gas-tight barrier rating)
X-link	Crosslinked (gel)

References

- Sanjay Vitthal, Aykut Goncu, Virginia Spencer, Ian Jackson, Xiaohui Xiao, Christy Y. Song, Mike Vanee, and Brenton Cheeseman, 2024. "Development and Qualification of a High Angle Multi-Frac (HAMF) Completion System for Deepwater Subsea Wells" paper presented at SPE ATCE in New Orleans, Louisiana, USA. SPE-220794-MS
- Anastasia Bird, Alistair Roy, Declan Tuffy, Alberto Casero, and James Hoad. "Next Steps in Hydraulic Fracturing in the Clair Field – An Evolving Story". 2023 Presented at ADIPEC in Abu Dhabi, UAE. SPE-216632-MS
- Shaun Thomson, Baglan Kiyabayev, Barry Ritchie, Jakob Monberg, Maurits De Heer, and Soren Skov List, 2022. "World's First Offshore Horizontal Well Using Jointed Pipe Cemented Frac Sleeve Technology". Presented at the SPE International Hydraulic Fracturing Technology Conference & Exhibition, Muscat, Oman. SPE-205331-MS
- Azwan Keong, Nelson Jaimes, Adlet Mussenov, and Hector Graterol, SLB; Bjørn Østebø, Max Sørensen, Karsten Maj, and Yann Caline, 2023. "World's Deepest Single-Trip Multistage Fracturing through 2-7/8-in. 7,400m Coiled Tubing String: Method Development and Learnings from Norway" presented at SPE/ICoTA Well Intervention Conference and Exhibition held in The Woodlands, Texas, USA.

Dustin Young, Tom Rune Koløy, Neil Kennedy, Trond Tollefsen, Håvard Jøranson, Marius Berge-Skillingstad, Alexander Letichevskii and Kjetil Skrettingland, 2024 "Single-trip Multistage Proppant Fracturing System Reduces Operational Time and Cost to Enable Cost-Effective Production of Tight and Low-Permeability Reservoirs on the Norwegian Continental Shelf", presented at OTC Houston 2024, OTC-35380